Flashcard Quizzer: Test Execution Report

Repository	github.com/dipanjan-acn/ud-ainse-assignment
Commit tested	2e8b2ff (main)
Platform	macOS 26.7.1
Interpreter	Python 3.11.17
Date	2026-10-24

> **Overall: PASS**
> Automated tests: 204 of 204 passed. Quality gates: 5 of 5 passed (100% coverage). Manual scenarios: 17 of 17 passed.

1. Quality gates

Gate	Result
black --check	PASS
isort --check-only	PASS
flake8	PASS
mypy strict	PASS
coverage 100 percent	PASS

Coverage: TOTAL 487 0 88 0 100%

2. Automated test suite (pytest)

Test file	Passed	Failed	Result
test_ai_generator	30	0	PASS
test_cli	16	0	PASS
test_cli_settings	9	0	PASS
test_code_limits	31	0	PASS
test_config	17	0	PASS
test_display	14	0	PASS
test_export	5	0	PASS
test_flashcard_loader	26	0	PASS
test_input_limits	7	0	PASS
test_integration	4	0	PASS
test_observers	7	0	PASS
test_quiz_engine	15	0	PASS

test_quiz_modes	15	0	PASS
test_quiz_runner	8	0	PASS
Total	**204**	**0**	PASS

3. Manual scenarios (scripted)

Each scenario runs the app, then checks its exit code and output. Terminal scenarios run inside a pseudo-terminal, so colors and Ctrl+C behave as they do for a user. Results come from `scripts/manual_tty_checks.py`.

PASS TTY-01: Sequential session, one miss, colors on

Command: `python main.py --mode sequential --file data/glossary.json --stats`
Expected: Exit 0; 9 green and 1 red feedback line; summary of 10 attempts at 90.0%
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: 9 green feedback lines
- ✔ PASS: 1 red feedback line
- ✔ PASS: summary shows 10 attempts
- ✔ PASS: accuracy is 90.0%

PASS TTY-02: NO_COLOR environment variable disables colors

Command: `python main.py -f data/glossary.json`
Expected: Exit 0; no escape codes in the output
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: no escape codes

PASS TTY-03: --no-color flag disables colors

Command: `python main.py -f data/glossary.json --no-color`
Expected: Exit 0; no escape codes in the output
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: no escape codes

PASS TTY-04: Adaptive session repeats a missed card

Command: `python main.py -m adaptive -f data/python_basics.json --stats --no-color`
Expected: Exit 0; 9 attempts for 8 cards, 8 correct
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: 9 attempts
- ✔ PASS: 8 correct

PASS **TTY-05: Typing 'exit' stops the quiz early**

Command: `python main.py -f data/glossary.json --no-color`
Expected: Exit 0; 'Leaving the quiz early.' is printed
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: early-exit message

PASS **TTY-06: Real Ctrl+C stops gracefully**

Command: `python main.py -f data/glossary.json`
Expected: Exit 0; 'Quiz stopped. Goodbye!' printed; no traceback
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: goodbye message
- ✔ PASS: no traceback

PASS **CLI-01: --help lists every flag**

Command: `python main.py --help`
Expected: Exit 0; help text names all six flags
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: lists --file
- ✔ PASS: lists --mode
- ✔ PASS: lists --stats
- ✔ PASS: lists --export
- ✔ PASS: lists --log-file
- ✔ PASS: lists --no-color

PASS **CLI-02: Card without 'back' gives a one-line error**

Command: `python main.py -f /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/missing_back.json`
Expected: Exit 1; 'Error:' names the field; no traceback
Exit code: 1

- ✔ PASS: exit code is 1
- ✔ PASS: names 'back'
- ✔ PASS: no traceback

PASS **CLI-03: Malformed JSON gives a friendly error**

Command: `python main.py -f /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/broken.json`
Expected: Exit 1; 'Invalid JSON' with a line number
Exit code: 1

- ✔ PASS: exit code is 1
- ✔ PASS: invalid JSON message
- ✔ PASS: no traceback

PASS **CLI-04: Missing deck file gives a friendly error**

Command: `python main.py -f /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/no_such_deck.json`
Expected: Exit 1; 'File not found' message
Exit code: 1

- ✔ PASS: exit code is 1
- ✔ PASS: file-not-found message

PASS **CLI-05: Export to a folder fails before the quiz starts**

Command: `python main.py -f data/glossary.json --export /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35`
Expected: Exit 1; error printed; no question is asked
Exit code: 1

- ✔ PASS: exit code is 1
- ✔ PASS: error message
- ✔ PASS: no question asked

PASS **CLI-06: Adaptive session, all cards correct (piped input)**

Command: `python main.py -m adaptive -f data/python_basics.json --no-color --stats`
Expected: Exit 0; 100.0% accuracy
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: accuracy is 100.0%

PASS **CLI-07: Export and log file written during a session**

Command: `python main.py -m adaptive -f data/python_basics.json --no-color --export /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/report.json --log-file /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/answers.log`
Expected: Exit 0; report has 2 attempts; log has 2 lines
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: report has 2 attempts
- ✔ PASS: log has 2 lines

PASS **CLI-08: Success-criteria command: sequential mode, glossary deck**

Command: `python main.py --mode sequential --file data/glossary.json`
Expected: Exit 0; quiz starts and exits cleanly
Exit code: 0

- ✔ PASS: exit code is 0
- ✔ PASS: first card shown

PASS CLI-09: Invalid mode in FLASHCARD_MODE gives a clear error

Command: `python main.py -f data/glossary.json`
Expected: Exit 1; names the unknown mode
Exit code: 1

- ✔ PASS: exit code is 1
- ✔ PASS: names the mode

PASS CLI-10: --generate without an API key stops before any network call

Command: `python main.py --generate networking --save /var/folders/m3/p7d1bn4x6zbfmtwnlftjg0n00000gp/T/tmp8tq8ml35/generated.json`
Expected: Exit 1; message names ANTHROPIC_API_KEY; no deck written
Exit code: 1

- ✔ PASS: exit code is 1
- ✔ PASS: names the variable
- ✔ PASS: no deck written

PASS CLI-11: --generate without --save is refused

Command: `python main.py --generate networking`
Expected: Exit 1; asks for --save
Exit code: 1

- ✔ PASS: exit code is 1
- ✔ PASS: asks for --save

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4. Earlier one-off runs

Check	Result	Note
Real SIGINT sent to the running process (kill -INT)	PASS	Stopped with "Quiz stopped. Goodbye!", exit 0, no traceback
Export to a folder, first version (after the quiz)	FAIL	The error appeared only after the quiz had run. Fixed.
Export to a folder, after the fix	PASS	Now covered by CLI-05, which fails before the first question

5. Limitations

- Windows was not tested. Terminal scenarios are reported as SKIPPED there, and the color output has not been checked on a Windows console.
- The automated suite and the scripted scenarios run on this one commit. Earlier commits were not re-tested.